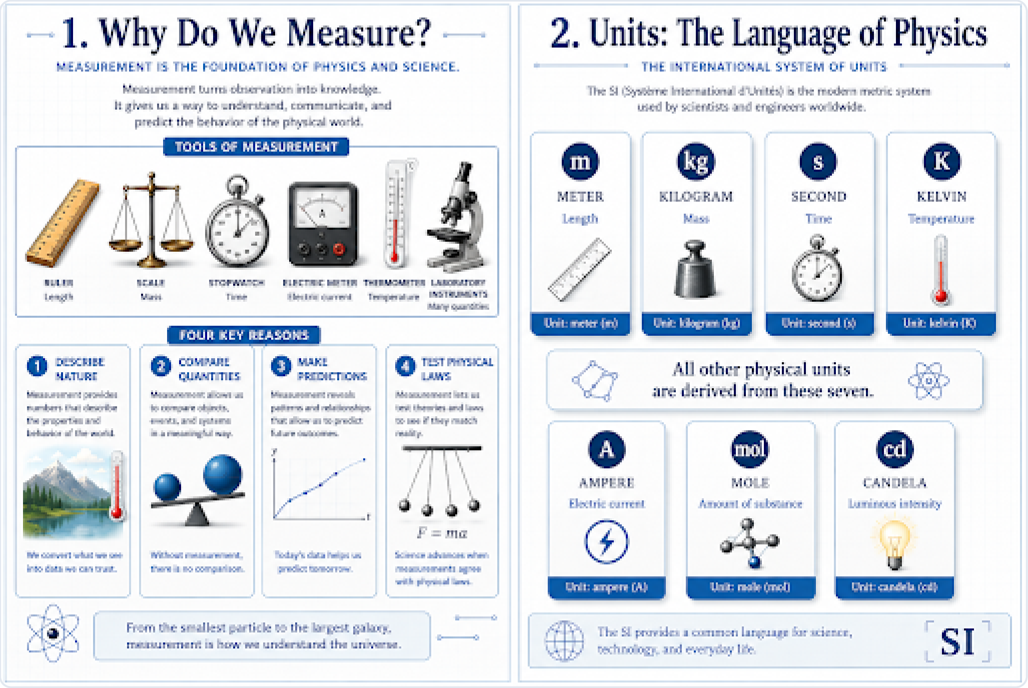
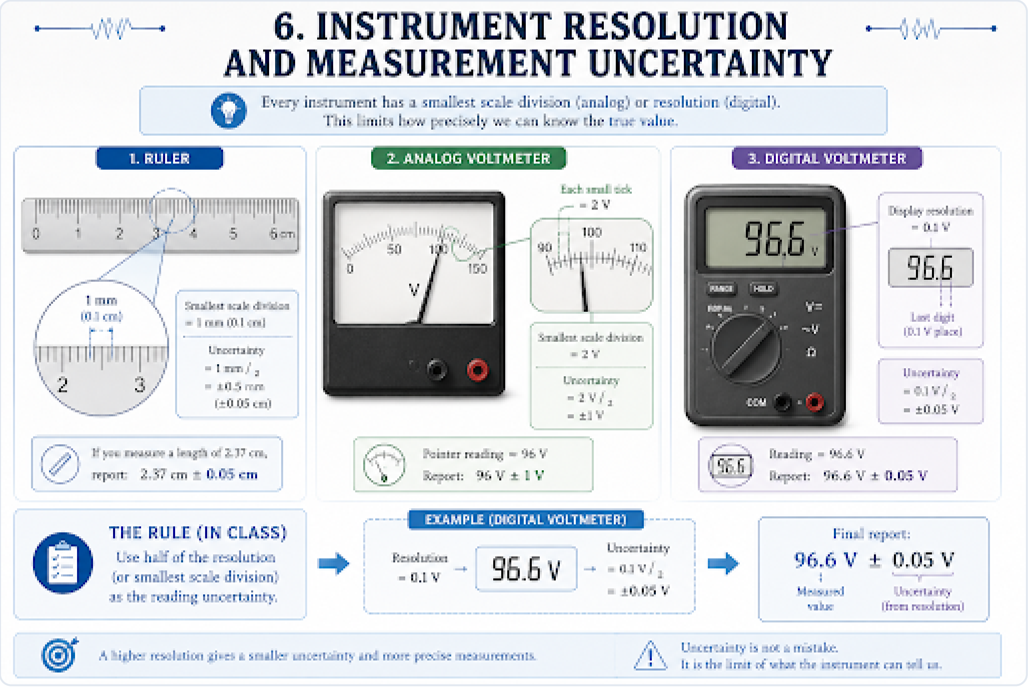




6. INSTRUMENT RESOLUTION AND MEASUREMENT UNCERTAINTY

Every instrument has a smallest scale division (analog) or resolution (digital).
This limits how precisely we can know the true value.

1. RULER

1 mm
(0.1 cm)

Smallest scale division
= 1 mm (0.1 cm)

Uncertainty
= 1 mm / 2
= ±0.5 mm
(±0.05 cm)

If you measure a length of 2.37 cm,
report: 2.37 cm ± **0.05 cm**

2. ANALOG VOLTMETER

Each small tick
= 2 V

Smallest scale division
= 2 V

Uncertainty
= 2 V / 2
= ±1 V

Pointer reading = 96 V
Report: 96 V ± **1 V**

3. DIGITAL VOLTMETER

Display resolution
= 0.1 V

Last digit
(0.1 V place)

Uncertainty
= 0.1 V / 2
= ±0.05 V

Reading = 96.6 V
Report: 96.6 V ± **0.05 V**

THE RULE (IN CLASS)

Use half of the resolution
(or smallest scale division)
as the reading uncertainty.

EXAMPLE (DIGITAL VOLTMETER)

Resolution
= 0.1 V → 96.6 V → Uncertainty
= 0.1 V / 2
= ±0.05 V

Final report:

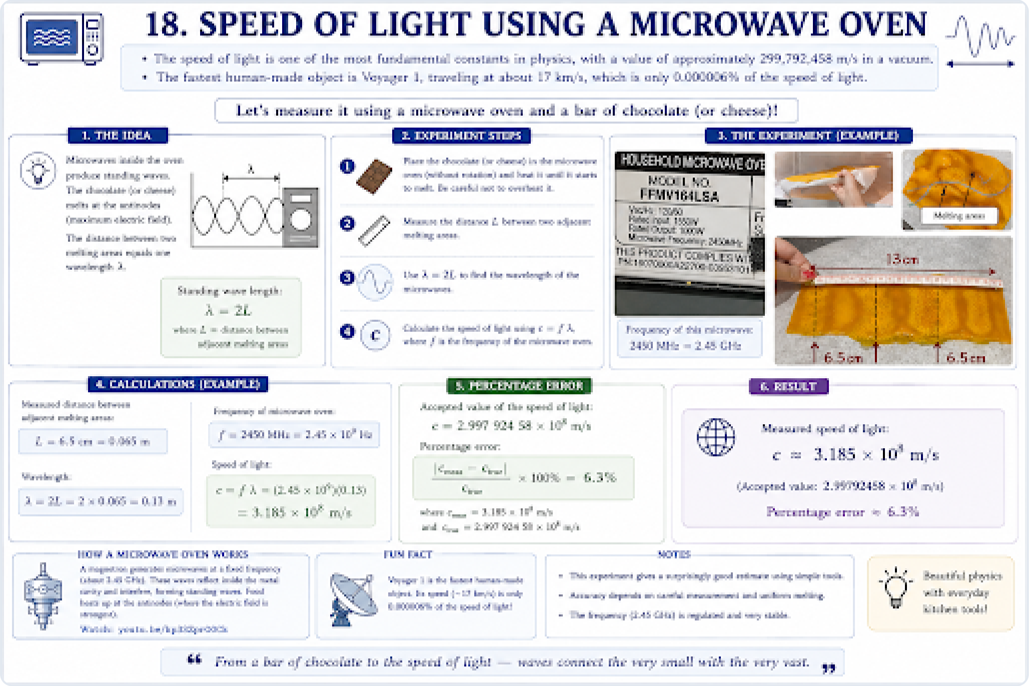
96.6 V ± 0.05 V

Measured value

Uncertainty (from resolution)

A higher resolution gives a smaller uncertainty and more precise measurements.

Uncertainty is not a mistake.
It is the limit of what the instrument can tell us.



18. SPEED OF LIGHT USING A MICROWAVE OVEN

- The speed of light is one of the most fundamental constants in physics, with a value of approximately 299,792,458 m/s in a vacuum.
- The fastest human-made object is Voyager 1, traveling at about 17 km/s, which is only 0.000006% of the speed of light.

Let's measure it using a microwave oven and a bar of chocolate (or cheese)!

1. THE IDEA

Microwaves inside the oven produce standing waves. The chocolate (or cheese) melts at the antinodes (maximum electric field).

The distance between two melting areas equals one wavelength λ.

Standing wave length:

$$\lambda = 2L$$

where L = distance between adjacent melting areas

2. EXPERIMENT STEPS

1. Place the chocolate (or cheese) in the microwave oven (without rotation) and heat it until it starts to melt. Be careful not to overheat it.
2. Measure the distance L between two adjacent melting areas.
3. Use $\lambda = 2L$ to find the wavelength of the microwaves.
4. Calculate the speed of light using $c = f\lambda$, where f is the frequency of the microwave oven.

3. THE EXPERIMENT (EXAMPLE)

Frequency of this microwave:
2450 MHz = 2.45 GHz

4. CALCULATIONS (EXAMPLE)

Measured distance between adjacent melting areas:

$$L = 6.5\ \text{cm} = 0.065\ \text{m}$$

Wavelength:

$$\lambda = 2L = 2 \times 0.065 = 0.13\ \text{m}$$

Frequency of microwave oven:

$$f = 2450\ \text{MHz} = 2.45 \times 10^9\ \text{Hz}$$

Speed of light:

$$c = f\lambda = (2.45 \times 10^9)(0.13) = 3.185 \times 10^8\ \text{m/s}$$

5. PERCENTAGE ERROR

Accepted value of the speed of light:

$$c = 2.997\,924\,58 \times 10^8\ \text{m/s}$$

Percentage error:

$$\frac{|c_{\text{meas}} - c_{\text{true}}|}{c_{\text{true}}} \times 100\% = 6.3\%$$

where $c_{\text{meas}} = 3.185 \times 10^8$ m/s
and $c_{\text{true}} = 2.997\,924\,58 \times 10^8$ m/s

6. RESULT

Measured speed of light:

$$c \approx 3.185 \times 10^8\ \text{m/s}$$

(Accepted value: 2.99792458×10^8 m/s)

Percentage error ≈ 6.3%

HOW A MICROWAVE OVEN WORKS

A magnetron generates microwaves at a fixed frequency (about 2.45 GHz). These waves reflect inside the metal cavity and interfere, forming standing waves. Food heats up at the antinodes (where the electric field is strongest).

Watch: youtu.be/kp3XQpvO0Ck

FUN FACT

Voyager 1 is the fastest human-made object. Its speed (~17 km/s) is only 0.000006% of the speed of light!

NOTES

- This experiment gives a surprisingly good estimate using simple tools.
- Accuracy depends on careful measurement and uniform melting.
- The frequency (2.45 GHz) is regulated and very stable.

Beautiful physics with everyday kitchen tools!

> *From a bar of chocolate to the speed of light — waves connect the very small with the very vast.*